Washington University in St. Louis

2026–2027 AIAA Design/Build/Fly (DBF) Competition Proposal

UAV with Sensor Suite — Aircraft: ***"Rocky"***

1 Executive Summary

The following proposal details Washington University in St. Louis Design/Build/Fly (WUDBF) team's plan to design, manufacture, and test its aircraft, *Rocky*, for the 2026–27 AIAA Design/Build/Fly Competition. The objective of this year is to design a UAV capable of delivering a towed sensor, transporting it in a protective shipping container, then deploying, operating, and recovering the sensor in flight, while also demonstrating the shipping container's ability to withstand multiple falls from a testing rig. *Rocky* is a high-wing, single-tractor monoplane with a conventional tail and tricycle landing gear, sized to a 5.9 ft wingspan (WS) to combine high-lift characteristics with efficient internal stowage of the sensor, shipping container, and winch mechanism. WUDBF's design strategy prioritizes minimizing overall weight and lap time for Mission 2 (M2), while prioritizing lap speed over sensor weight to maximize score in Mission 3 (M3). Initial performance targets are: an automatic Mission 1 (M1) score via a waived M1 attempt bundled into M2, 5 laps in M2 carrying the sensor in its container and 1 container simulator at a projected 5.06 lb combined weight at a total mission time of 178.7 s; 7 laps in M3 with a 2.2 lb sensor; and a Ground Mission (GM) staging/drop-test sequence from a height of 60 in at the 2.2 lb sensor weight, targets that reflect the scoring balance between the M2 weight÷time term and the M3 laps×weight term. The team's approach is built on an in-house MATLAB optimizer, trade studies used to select the airframe and sensor-deployment configuration, a manufacturing plan built around proven composite and prototyping techniques already validated in past competition years, and a test plan that validates the deployment, recovery, and structural systems. WUDBF's technical work is coordinated by the leadership team and includes risk, schedule, and budget management.

2 Management Summary

2.1 Organizational Structure

WUDBF is a student-run club with a faculty advisor, consisting of 83 active members across administrative and technical sub-teams (18 freshmen, 19 sophomores, 17 juniors, 15 seniors, and 14 dual-degrees). The executive board is split into administrative and technical branches to streamline operations, as seen in Figure 1. The administrative president oversees three leads: Business (budget management and sponsorship acquisition), Marketing (social media), and Graphic Design (branding and proposal/report formatting). The technical president oversees six sub-teams: Aerodynamics and Payload, Avionics, Propulsion, Structures, Manufacturing, and Simulation Development (Table 1). Sub-team leaders run weekly meetings and track deliverables against the deadlines of the schedule in Section 2.2. This structure, refined over the past three competition years, distributes decision-making authority among each lead, while the technical president coordinates work across each sub-team.

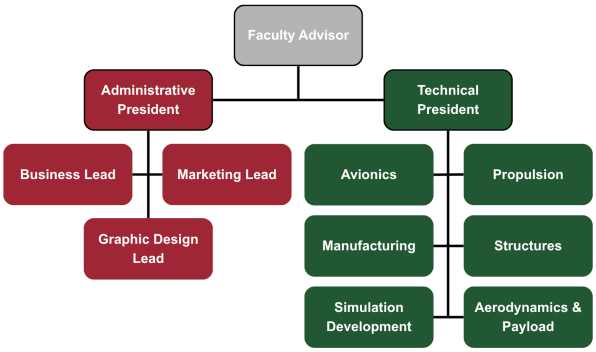


Figure 1: Team Organization Chart.

Table 1: Subteam Roles and Skills

Sub-team	Roles	Skills
Aerodynamics & Payload	Aerodynamic lift and drag analysis and aircraft sizing. Payload design and testing.	• Wind tunnel testing • SolidWorks CAD, XFLR5, and Ansys CFD
Avionics	Subcomponent wiring, sensor microcontroller programming, and avionics system installation. Test flight data collection.	• Soldering and crimping • Programming and control surface trimming
Manufacturing	Material selection and assembly of the prototype and competition aircraft.	• Resin infusion and machining • Composites, laser cutting, and 3D printing
Simulation Development	Optimizer coding and usage of simulation tools to set performance goals and run sensitivity analyses.	• MATLAB coding and test flight data analysis • Knowledge of competition scoring and rules
Propulsion	Motor, propeller, and battery selection and testing.	• Thrust-stand data analysis • Propulsion system analysis with eCalc/RC benchmark.
Structures	Analysis of aircraft stresses, design of aircraft CAD package, and internal structural layout determination.	• Solidworks CAD and FEA • Assembly and component design

2.2 Schedule

The team's projected schedule is shown in the Gantt chart in Figure 2. Sub-teams meet individually during the week, leadership meets Fridays to solve cross-team problems, and the full team meets Saturdays to stay aligned on deadlines.

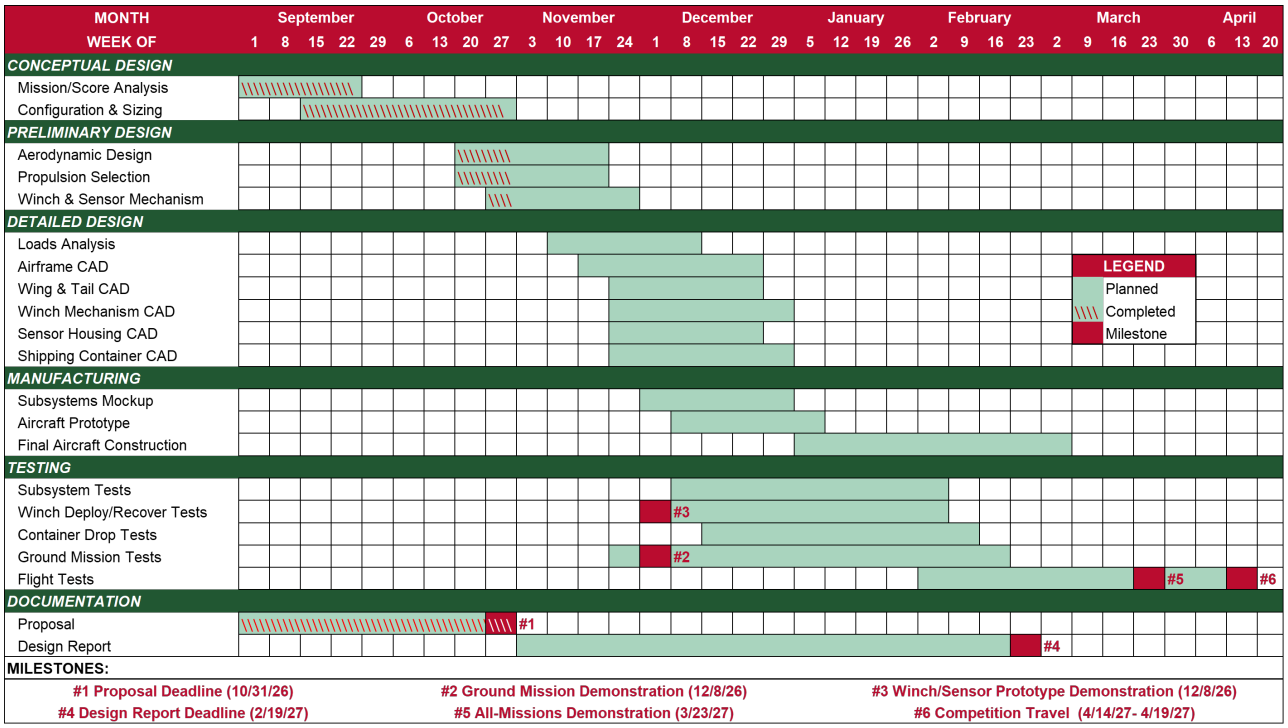


Figure 2: WUDBF 2026-27 schedule.

2.3 Budget

WUDBF receives funding from two primary sources: the Washington University Student Union and the McKelvey School of Engineering Undergraduate Student Services Grant. The budget in Table 2 contains supplies for this year's missions, the sensor, its shipping container, and the winch deployment/recovery mechanism, in addition to airframe materials, electronics, and competition travel. The budget was estimated from last year's expenses, fees, and the estimated cost of traveling to Tucson, Arizona. The contingency line covers a second shipping-container prototype, since the drop-test requirement carries the highest risk of requiring a design iteration after ground testing.

Table 2: 2026–2027 Budget.

Category	Item	Est. Cost
Manufacturing ($4,400)	3D Printing Materials	$100
	CNC Hot-Wire Cutting Supplies	$600
	Adhesives/Consumables	$500
	Carbon Fiber Spars/Tubes	$200
	Carbon Fiber Twill	$900
	Foam Core (Wing/Container)	$700
	Balsa & Birch (Wing/Tail Ribs)	$600
	PVC Resin Infusion Supplies	$800
Propulsion, Avionics, & Electronics ($3,400)	Motors	$700
	ESCs	$400
	Batteries	$700
	Propellers	$200
	Receivers/Transmitters	$300
	Winch Servo & Motor	$600
	Misc. Electronic Components	$500
Sensor & Container Systems ($2,600)	Sensor Housing Materials	$400
	Shipping Container Build Materials	$700
	Winch Spool/Reel Assembly	$700
	Sensor Light Electronics	$300
	Container Drop-Test Spare	$500
Flight Test ($250)	Miscellaneous/Field Consumables	$250
Travel ($11,100)	Lodging	$2,500
	Flight & Luggage	$6,600
	Car Rental	$500
	Fuel	$500
	Food	$1,000
Fees ($550)	Fly-Off Registration Fee	$550
Contingency ($2,000)	Spare Parts and Multiple Prototype Risk	$2,000
Total		$24,300

3 Conceptual Design

3.1 Mission Requirements Analysis

This year's competition consists of designing a surveillance UAV for three flight missions and one ground mission. The UAV must be able to deliver a sensor in a shipping container and may also deliver optional simulators (M2). The UAV must then deploy, tow, operate, and recover the sensor for surveillance via a tow line at least 1.5×WS (M3). Lastly, the GM stipulates testing the handling and drop-test survivability of the sensor shipping container. Table 3 interprets each mission requirement into subsystem targets that drive the design choices in Sections 3.2–3.5, mainly based on sensor weight (M3), # of laps flown (M3), lap speed (M2), payload gross weight (M2), and shipping container drop height (GM).

Table 3: Mission Overview.

Mission	Score for Mission	Mission Requirements	Subsystem Requirements
M1: Staging Flight	1.0	• Prepare aircraft for flight in under 5 minutes • Complete 3 laps within 5 min window • Successful landing required for score • Maximum aircraft wingspan & weight: 6 ft & 55 lb	• Stable flight configuration required • Hatches allow easy access to subsystems • Aircraft must be easy to configure prior to the staging window

M2: Delivery Flight	$1 + \frac{\left[\frac{\text{Weight}}{\text{Time}}\right]_{\text{Team}}}{\left[\frac{\text{Weight}}{\text{Time}}\right]_{\text{Max}}}$	• Carry sensor at declared weight in shipping container (+ optional simulators) • Complete 5 laps within 5 min window • Scored on (payload weight)/(flight time) • Each simulator must match the container's shape/size (±1/8 in.) and weigh the same as the container + declared sensor weight (±1 oz).	• Container bay sized for sensor + container + simulator(s) • CG held within design limits at full payload • Secure, rigid payload restraint (no in-flight shift) • Fuselage and wings must be sturdy enough to support a heavy aircraft gross weight
M3: Sensor Flight	$2 + \frac{[\#\text{ Laps} \times \text{Sensor Weight}]_{\text{Team}}}{[\#\text{ Laps} \times \text{Sensor Weight}]_{\text{Max}}}$	• Deploy sensor after takeoff, before 1st turn (lights OFF) • Lights SOLID ON after lap 1, FLASHING ON after lap 2 • Stow sensor (lights OFF) before landing • Scored on (# laps) × (sensor weight) • Winch deploys sensor ≥ 1.5×WS	• High airframe tensile strength for towing • Sensor lighting must be distinct and bright • Reliable and stable stow/recovery mechanism • Sensor shape must be optimized for aerodynamic stability and low drag
GM: Ground Mission	$0.5 + \frac{[\text{Weight} \times \text{Height}]_{\text{Team}}}{[\text{Weight} \times \text{Height}]_{\text{Max}}}$	• Sensor at max Tech-Inspection weight • 6 drops (2/side, 3 sides) at declared height (≤60 in.) • Full rigging/handling cycle within 5 min limit	• Container must survive repeated drops without damage to sensor • Minimize rigging installation time • Rigid, shock-absorbent container structure
Total Mission Score = M1 + M2 + M3 + GM			

3.2 Scoring Optimization and Sensitivity Analysis

A MATLAB optimizer was created from scratch to size the aircraft and sensor system, using the scoring equations for M2, M3, and GM as the objective function and wing area, cruise velocity, sensor weight, motor power, plane empty weight, container weight, and number of simulators as design variables. The optimizer was benchmarked against historical top-three performance data to set realistic sizing margins. To establish a feasible design space for plane sizing, a constraint diagram was built by sweeping aspect ratio and cruise velocity against a required thrust-to-weight ratio and stall speed, as seen in Figure 3. The optimizer was constrained to search only within this region rather than the full design space, and the selected geometry, aspect ratio, and thrust/weight ratio were selected with margin to the region boundary. A sensitivity analysis was then conducted using the optimizer to guide the conceptual design process, as seen in Figure 4. The sensitivity study sweeps each design variable ±20% from its optimized value and reports the resulting fractional change in total mission score (M2 + M3 + GM). GM drop height was set at the maximum height of 60 in. Beyond the variables entering the scoring equations directly, the study also includes wing area, cruise power, empty weight (with avionics and propulsion systems), and number of simulators, each of which constrains the design space without appearing in the scoring formulas themselves. Sensor weight shows the steepest slope, consistent with its multiplicative roles in the M3 and GM scores, respectively; a 20% change from the optimized baseline moves total score by 8–15%. Cruise velocity and number of simulators show comparable slopes, contributing secondarily through the M2 scoring term. Wing area is the only variable showing non-monotonic behavior, falling sharply above roughly 10% over baseline. Cruise power shows effectively zero sensitivity, confirming the 552.4 W power was sized with margin above the M2/M3 thrust minimum. The final optimized design values from the script are given in Table 4. To validate that the balanced objective weighting above was appropriate rather than biased toward one mission, the optimizer was re-run under two alternative extreme weightings (Table 5), a speed-maximized weighting and a payload-maximized weighting, and compared against the balanced convergence point in Table 4. Each configuration utilized a 0.33 lb shipping container and a drop height of 60 in. The total score was normalized with the highest performer in each mission. The speed-focused approach vastly underperformed the other approaches, while the payload-focused scored similarly, but ultimately scored lower due to its longer lap time.

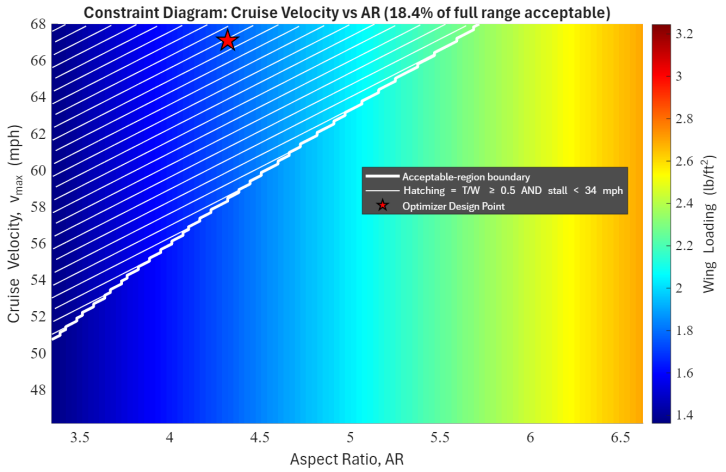


Figure 3: Constraint diagram.

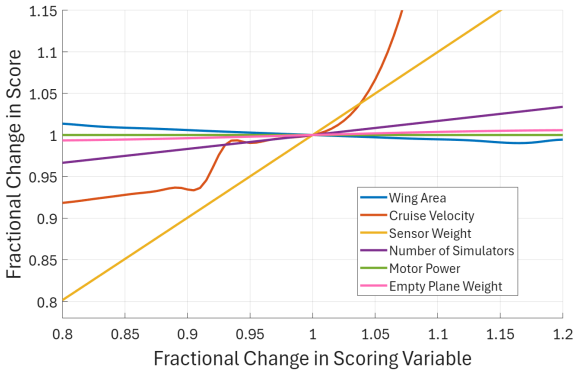


Figure 4: Sensitivity Analysis.

Table 4: Optimizer Values.

Parameter	Value
Wing Area	8.06 ft²
M3 Laps	7
M2 Time	178.7 s
Target Sensor Weight	2.2 lb
Number of Simulators	1
Plane Empty Weight	7.74 lb
Required Cruise Power	552.4 W
GM Drop Height	60 in
Container Weight	0.33 lb

Table 5: Objective-Function Weighting Validation

Configuration	Sensor Wt.	# of Simulators	Wing Area	Cruise Speed	AR	M3 Laps	M2 Wt. ÷ Time	M3 Laps × Sensor Wt.	GM Drop Height × Wt.	Total Score
Speed-Focused	0.55 lb	0	4.72 ft²	114.84 ft/s	7.52	10	0.006	5.5	33	**0.78**
Payload-Focused	2.00 lb	3	16.09 ft²	66.15 ft/s	2.16	6	0.035	12	120	**2.69**
Balanced	2.20 lb	1	8.06 ft²	98.41 ft/s	4.32	7	0.028	15.4	132	**2.80**

3.3 Design Approach

Based on the results of the scoring optimization, WUDBF targets a configuration sized for a balance of speed, payload capacity, and sensor weight, rather than prioritizing one scoring element over the others. Per this year's rules, teams may waive an M1 attempt and proceed directly to M2, with a successful M2 awarding a full M1 score. WUDBF will bypass M1 in favor of M2, as this saves a flight attempt for a re-flight of M2 or M3 if needed. Historically, the team has completed sufficient flight testing to mitigate risk associated with waiving M1. For M2, the aircraft will carry the sensor in its shipping container plus one container simulator, targeting 5 laps at 178.7 s to maximize the weight/time scoring term. For M3, the sensor will be deployed before the first downwind 180 degree turn, cycled through its three light modes as specified in the rules, and recovered before landing. The winch will deploy the sensor to 9 ft, with a target of 7 laps at the 2.2 lb sensor weight identified in the sensitivity study. For the GM, the team targets a drop height of 60 in. and completion of the full GM in under 3 min 30 s, leaving margin against the 5-minute limit.

3.4 Trade Studies

Trade studies were conducted using a weighted decision matrix for the six configuration decisions with the largest impact on score. Each competing design was graded by categories determined from subsystem requirements. Each matrix was re-scored with ±15% weighting perturbations to eliminate any biases and confirm the selected configuration remained the top choice. As seen in Table 6, a high-wing design was selected decisively over low-wing and mid-wing, driven primarily by its superior ground clearance for loading payload and better stability. A monoplane design was chosen decisively over twin boom and flying wing, scoring highest on every single criterion, reflecting the team's existing manufacturing and design familiarity with the configuration. Tricycle landing gear was selected over taildragger for its advantage in clearance and stability, especially considering this year's missions. As seen in Table 7, the conventional tail outscored the alternatives, driven primarily by its manufacturability advantage and structural stability during flight. A single-tractor layout was chosen over both twin tractor and single pusher: a pusher configuration places the propeller close to the trailing tow line and deployed sensor, creating a risk of the sensor hitting it during recovery, and the resulting low propeller clearance score

Table 6: Wing Location, Aircraft Configuration, and Landing Gear Trade Studies.

	Scoring Weight	High Wing	Low Wing	Mid Wing		Scoring Weight	Monoplane	Flying Wing	Twin Boom		Scoring Weight	Tricycle	Taildragger
Ground Clearance for Winch/Sensor	30%	9	4	6	Payload Bay Volume	25%	8	6	7	Ground Handling & Visibility	25%	9	5
Stability (Dihedral Effect)	20%	8	4	6	Stability & Control	25%	8	5	7	Container Loading Ground Clearance	25%	8	5
Payload/Container Access	20%	8	9	6	Tow-Line/Winch Clearance	20%	8	6	9	Structural Weight	20%	6	8
Structural Integration	15%	6	8	5	Manufacturing Complexity	15%	8	4	6	Crosswind Stability	15%	8	5
Manufacturability	15%	7	8	6	Team Experience	15%	9	2	3	Manufacturability	15%	7	6
Weighted Result		7.85	6.20	5.85	Weighted Result		8.15	4.85	6.65	Weighted Result		7.70	5.75

Table 7: Tail Configuration, Propulsion Configuration, and Sensor Deployment Trade Studies.

	Scoring Weight	T-Tail	Conventional Tail	V-Tail		Scoring Weight	Single Tractor	Twin Tractor	Single Pusher		Scoring Weight	Bottom	Rear
Points of Failure	15%	5	8	5	Propeller Clearance from Tow Line	30%	9	7	5	Deployment & Recovery Reliability	30%	6	7
Yaw Authority	30%	7	7	8	Thrust Efficiency at Cruise	25%	8	6	7	Aerodynamic Interference	20%	8	5
CG Effects	15%	6	7	7	Wake Effects	20%	6	7	8	CG Effects	20%	8	4
Wake Effects from Wing	15%	8	6	7	Team Experience	15%	10	6	4	Winch Integration Complexity	15%	8	6
Manufacturability	25%	5	9	4	Weight	10%	7	5	7	Ground Clearance During Staging	15%	5	8
Weighted Result		6.20	7.50	6.25	Weighted Result		8.10	6.40	6.15	Weighted Result		6.95	6.00

was enough to separate it from single tractor regardless of weighting changes. The single tractor was chosen over a twin tractor because it was more efficient with power and weight. Bottom-mounted sensor deployment was selected over a rear-mounted location, driven primarily by its advantage in aerodynamic interference and center of gravity (CG) effects.

3.5 Preliminary Design

Upon receiving the optimized design values and chosen configurations, the sub-teams proceeded with the preliminary aircraft design. A CAD rendering of the preliminary aircraft is shown in Figure 5. XFLR5 software was utilized to compare several candidate airfoils, prioritizing low drag, maneuverability, and a stable moment coefficient. The Drela GW-27 airfoil was chosen due to its superiority in the aforementioned qualities and high lift/drag ratio at cruise Reynolds numbers for the M2 and M3 configurations. The airfoil was then modified using an in-house airfoil design software to be rounder at the leading edge and with a slightly larger camber to increase lift for performance optimization. The wing planform area was set at 8.06 ft^2, a value represented in the constraint diagram as inside the feasible region for thrust-to-weight and stall speed with margin, a balanced selection rather than being selected deeper into the low wing-loading corner where M2 weight/time performance would suffer. The wing planform features a dihedral angle of 4° for overall aerodynamic stability, especially when towing the sensor, and winglets will be integrated into the wing to reduce the effects of wingtip vortices. Tricycle landing gear was integrated per the landing gear trade study (Section 3.4). Utilizing MotoCalc and eCalc software, a propulsion system was selected that met performance requirements; the Scorpion SII-4035-380 Kv was selected for its ability to exceed the thrust-to-weight ratio of 0.5 identified in the constraint diagram while providing extra thrust margin to overcome high wind gusts and sustained winds that may occur during competition. Preliminary aircraft parameters are provided in Table 8. For shipping container design (Figure 6), an exterior consisting of aluminum was chosen to protect the sensor during GM falls, with an inner layer of rubber and a foam core for maximum shock absorbance. The rigging system was also integrated into the container with attachments to the rubber layer. The simulator was designed to be wooden with a metal core to replicate the size and weight of the shipping container with the sensor, per the rules requirement that each simulator match the container's shape and size within ±1/8 in. and its loaded weight within ±1 oz. The fuselage was designed to load the simulator through a top door, behind the propulsion system, while the shipping container is to be loaded from a rear hatch, since the container cannot clear the wing spars if loaded from the top. For M3, the sensor mechanism was designed to be directly underneath the wings, per the sensor deployment location trade study (Section 3.4), with the sensor CG matched to the aircraft CG. A winch mechanism was chosen to deploy and retract the sensor with two attached cables, extending to 9 ft. Two cables were chosen to reduce the tensile force on the mechanism and fuselage structure during M3 flight. Two automatic doors on the belly of the aircraft open during M3. A motor powers two spools that connect the cables to the sensor, and the cable further aft contains the electrical connection to control the LEDs. Preliminary wind tunnel testing was conducted to evaluate four candidate fin arrangements for the towed sensor: cruciform, three-fin-up, three-fin-down, and dihedral-mounted. Each configuration was held by a servo at an initial 5° angle inside the test section before being let go, with stability assessed directly from the measured response using settling time and RMS angular deviation (Figure 7).

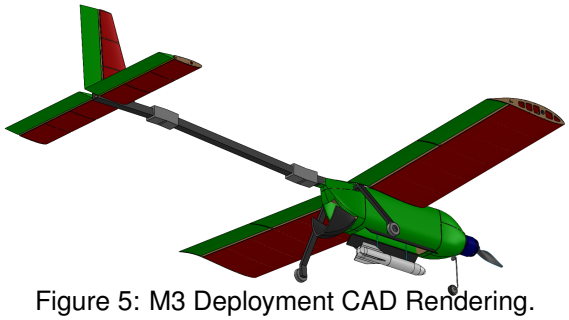


Figure 5: M3 Deployment CAD Rendering.

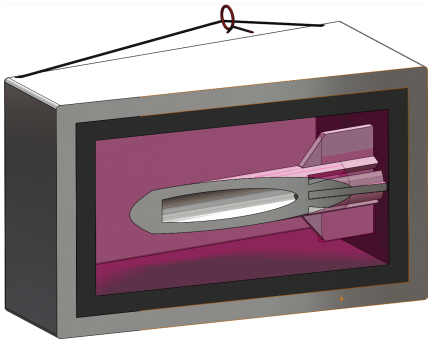


Figure 6: Shipping Container CAD Cross-Sectional View.

Table 8: Aircraft Parameters.

Propulsion System		Aircraft Sizing	
Propeller	APC 14x8.5	**Wingspan**	5.9 ft
Motor	Scorpion SII-4035-380KV	**Wing Chord**	16.4 in
Battery	8S 3300 mAh	**Aspect Ratio**	4.32
Thrust/Weight	1.00	**Airfoil**	GW-27 (mod)
Maximum Thrust	12.76 lb	**Wing Loading**	1.59 lb/ft^2
Battery Capacity	97.7 Wh	**Max TOGW**	12.8 lb

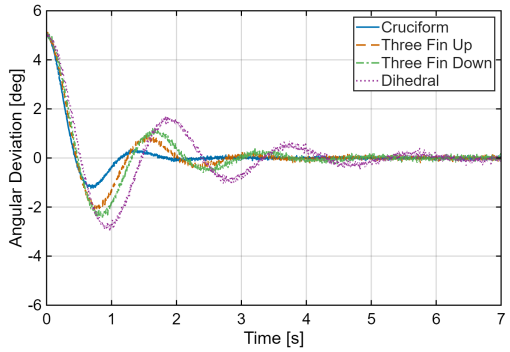


Figure 7: Sensor Fin Analysis.

The cruciform configuration settled fastest (5.96 s) and exhibited the lowest RMS deviation (0.717°). Thus, the sensor will contain four cruciform-arranged aluminum fins for aerodynamic stability. The sensor will be made of aluminum, chosen for the added mass it contributes toward the M3 score relative to a lighter composite alternative, while also providing sufficient strength to survive repeated deployment and recovery cycles. Lastly, the sensor will be powered by a 7.6V 2S LiPo battery and utilize an A-Star 32U4 Micro microcontroller board, chosen because of their compact sizes.

4 Manufacturing Plan

The manufacturing plan follows the design, manufacturing, and testing workflow shown in Figure 8. Following Detailed Design, the aircraft is developed along two parallel tracks: a design-validation track and a manufacturing-readiness track, which converge into a single Complete Design before full-scale fabrication begins. Purchased components (the propulsion system, servos, receiver, and winch motor) are integrated during the avionics/propulsion procurement branch, while the fuselage, wing/tail structures, sensor housing, winch bracket, and shipping container are fabricated in-house. The fuselage will use a 2-ply carbon fiber with PVC foam resin infusion layup over a foam plug, chosen for its strength-to-weight ratio and the team's existing experience with the process, and is fabricated alongside birch airframe manufacturing and subsystem integration. Wing and tail structures will use balsa and birch ribs over carbon fiber spars, covered in MonoKote, selected for easy manufacturability and repairability during the flight-test phase, with wing and tail assembly proceeding in parallel before integration with the fuselage. The sensor housing and winch mounting bracket will be machined out of 6061 aluminum for durability and strength and integrated alongside avionics and propulsion procurement and testing. The shipping container will have a machined 6061 aluminum casing with EPDM rubber and cross-linked polyethylene foam inner layers. The three parallel subassembly branches converge at Final Assembly and Testing.

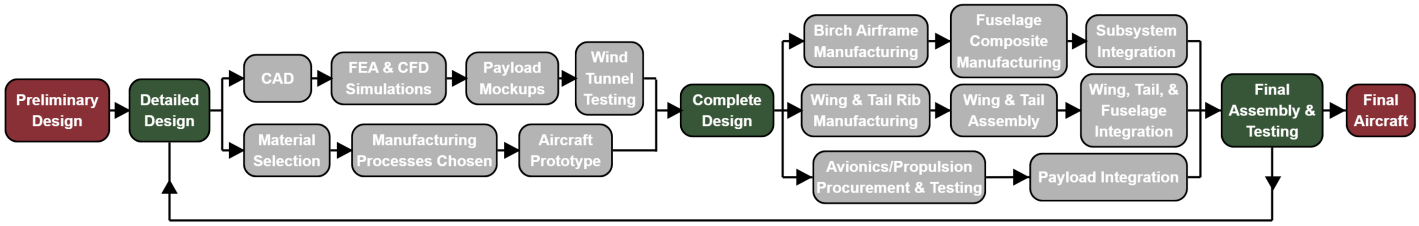


Figure 8: Design, Manufacturing, and Testing Workflow.

5 Test Planning

Table 9 outlines the static and dynamic test plan. Priority is given to validating the two subsystems that are new this year: the container drop test and the winch & sensor prototype flight are treated as gating tests, since neither the shipping container nor the winch mechanism will be integrated onto the final competition airframe until their respective tests pass. The test plan feeds data back into the design loop rather than only confirming requirements. Static tests refine each subsystem's predicted performance and safety margins before integration, while dynamic tests validate those predictions in flight using pilot feedback, telemetry, and propulsion data to inform trim and design refinement ahead of the next attempt.

Table 9: Test Plan

Sub-team	Test	Objective	Method	Due Date
Static Testing				
Propulsion	Static Thrust Test	Verify candidate motor/propeller/battery combinations achieve ≥12.76 lb static thrust at <95 A and within ±10% of eCalc predictions.	Use a motor test stand to measure static thrust, current draw, and efficiency across candidate propeller/battery combinations.	11/3/26
Structures	Structural Test	Verify wing, fuselage, and gear withstand 2.5g equivalent wingtip lift at 12.8 lb MTOW with <1.5 in. tip deflection and zero permanent set.	Support the aircraft by its wingtips at 1.25x MTOW (about a 3.1g flight load) to verify a 1.25 safety factor over tech inspection, and drop-test the gear.	12/8/26
Structures	CG Test	Verify aircraft CG remains within ±0.25 in. of target 25% MAC (8–12% static margin) across M1/M2/M3 payload states and full sensor deployment.	Suspend the aircraft from its wingtips at M2 and M3 payload configurations and measure CG location.	12/8/26
Full Team	GM Rehearsal Test	Minimize the shipping container rigging/handling time for the Ground Mission.	Execute timed trials of all 3 rigging swaps and container unlatching until 5 consecutive runs finish in ≤3 min 30 s	12/8/26
Structures	Winch Mount Load Test	Verify the winch mounting bracket and sensor attachment point withstand loading during deployment and recovery.	Apply 1.5x max predicted tow-line tension to winch bracket, spool, and dual cables for 5 min with zero bracket yielding or cable slip.	12/11/26
Manufacturing	Container Drop Test	Verify container survives 6 drops onto concrete from 60 in. with zero sensor structural denting/buckling, intact rigging mounts, and 100% LED function.	Drop the container from the declared 60 in. height on all three required sides, two drops per side, and inspect for structural damage.	1/5/27
Avionics	Electronics & Light-Mode Test	Ensure all electronics function correctly, including receiver-transmitter connection at range and the three sensor light modes (OFF/SOLID/FLASHING).	Verify fail-safe surfaces and command OFF/SOLID/FLASHING light modes over the 9 ft tow cable in range-check mode with 100% daytime visibility at 500+ ft.	1/12/27
Dynamic Testing				
Avionics	Winch & Sensor Prototype Flight	Verify in-flight deployment/recovery to ≥9 ft in less than 12 s and confirm towed pitch/yaw oscillation stays within ±2° with <15% elevator trim shift.	Fly a prototype aircraft, deploy and recover the sensor in flight, and observe drag/trim response.	12/8/26
Full Team	First Flight	Verify safe takeoff in <100 ft, stable cruise at 98.4 ft/s, stall speed ≤34 mph (49.9 ft/s), and repeatable spot landings with <10% control surface trim.	Fly the aircraft unloaded and observe structural response and handling qualities.	1/26/27
Full Team	M2 Test	Complete 5 laps carrying the 5.06 lb payload (loaded container + 1 simulator) in ≤178.7 s while consuming ≤85 Wh (<87% of the 97.7 Wh battery capacity).	Fly the aircraft carrying the sensor, container, and simulator at the declared payload weight; measure lap time and implement improvements in subsequent tests.	2/9/27
Full Team	M3 Test	Complete ≥7 laps within the 5-min window towing the 2.2 lb sensor, verifying deployment before Turn 1, mode transitions on Laps 1–2, and full retraction.	Fly the full M3 sequence and compare lap count and sensor-weight performance to the sensitivity study's predicted targets.	3/2/27
Full Team	All-Missions Demonstration	Demonstrate the complete M2, M3, and GM sequence to confirm readiness for the fly-off.	Complete 5-min staging box assembly and back-to-back M2, M3, and GM runs in competition order with zero hardware failures.	3/23/27